

A localized mass bound and weighted maximum principle

Conditional analytic theorem; three independent mathematical reviews completed

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This note isolates the measure-theoretic final step of the proposed endpoint-profile improvement. It does not assert that all its kernel comparison hypotheses have already been proved for the dice tensor. All intervals below are fixed compact subintervals of $(0, \infty)$. Write $b = \sqrt{\tau}$, where $\tau = C_0 \log m/m$, and let J_0 lie strictly inside the interior of $J = [a_-, a_+]$.

We use positive kernels $K_t(x, a)$, for $t = \tau, c\tau$ with fixed $c > 1$, which on J satisfy fixed Gaussian upper bounds

$$K_t(x, a) \leq Cb^{-1}e^{-|x-a|^2/(Cb^2)}, \quad K_\tau(x, a) \geq c_1b^{-1} \quad (|x-a| \leq b).$$

The argument is unchanged if all distances are in $\sqrt{x}$, since that coordinate is uniformly bi-Lipschitz on J . All measures have mass bounded by a fixed constant. Kernel integrals outside J , for centers at distance δ from its boundary, are assumed to be at most $Cm^Cb^{-1}\exp[-c\delta^2/b^2]$; the same estimate is assumed after multiplication by any auxiliary functions used below. This can instead be supplied with exponential-moment weights. Suppose a reference measure β has a density bounded above and below by positive constants near J , so $c \leq \beta(K_\tau(\cdot, a)) \leq C$ for centers separated from the boundary by much more than b .

Lemma 1 (Absorbing a broader localized kernel). *Suppose a positive measure μ satisfies, uniformly for $a \in J$,*

$$|\mu(K_\tau(\cdot, a)) - \beta(K_\tau(\cdot, a))| \leq \epsilon_m \mu(K_{c\tau}(\cdot, a)) + m^{-B}, \quad \epsilon_m \longrightarrow 0,$$

where B can be any sufficiently large fixed constant. Then uniformly for $a \in J_0$,

$$\mu([a-b, a+b]) \leq Cb, \quad c \leq \mu(K_\tau(\cdot, a)) \leq C.$$

Consequently, for $a \in J_0$ and $h \geq b$ whose h -neighborhood remains in a fixed interior subinterval of J ,

$$\int_{|x-a| \geq h} K_\tau(x, a) d\mu(x) \leq Ce^{-ch^2/b^2} + O(m^{-B}).$$

Proof. Let $d(a) = \min(a-a_-, a_+-a)$, and use the weight $\phi(a) = d(a)^2$. Set $Q = \sup_{a \in J} \phi(a)\mu([a-b, a+b])/b$. It suffices to treat $Q \geq 1$. Choose an approximate maximizer a with value at least $Q/2$. The total-mass bound gives $\phi(a) \geq cQb \geq cb$, so its boundary distance $\delta = d(a)$ satisfies $\delta \geq c\sqrt{b} \gg b$. On $|x-a| \leq \delta/2$, the weights $\phi(x)$ and $\phi(a)$ are comparable. Partition that neighborhood into intervals of length b ; the Gaussian upper bound and the definition of Q give

$$\mu(K_{c\tau}(\cdot, a)) \leq CQ/\phi(a) + Cm^Cb^{-1}e^{-c\delta^2/b^2}.$$

The second term is superpolynomially small because $\delta^2 \geq cb$. The lower kernel bound gives $\mu(K_\tau(\cdot, a)) \geq cQ/\phi(a)$. Multiply the comparison hypothesis by $\phi(a)$ to obtain $Q \leq C + C\epsilon_m Q + o(1)$. Absorption proves $Q \leq C$. This gives the local interval bound on every fixed interior subinterval. Summing Gaussian tails over the same length- b partition proves the last assertion and the uniform upper bound on broader kernel masses. The comparison hypothesis and the positive reference integral now give the positive lower bound. $\square$

Theorem 1 (Weighted maximum principle). *Suppose there are finitely many marked row systems with positive measures μ_i , scalar coordinates x_i , and functions $h_i = z_i - x_i$, where z_i is nondecreasing as a function of the row coordinate x_i . Flat rows are allowed, with the corresponding order interpretation. Suppose the local-mass and kernel-mass conclusions of Lemma 1 hold uniformly for these measures for all centers in J . This follows by applying that lemma on a slightly larger fixed interval, with J as its interior interval. Assume a prior polynomial bound*

$$|h_i| \leq R_m \leq m^{-\alpha}, \quad \alpha > 0,$$

on J , after weakening a power slightly to absorb logarithmic factors. Let $H_i(x)$ control the row range occurring in the marked error. For the weight $\phi(x) = d(x)$, assume that, if $D = \max_{i, \text{rows}} \phi(x_i)|h_i|$ and $a \in J$, then

$$H_i(x) \leq CD/\phi(a) \quad \text{whenever } |x_i - a| \leq d(a)/2. \quad (\text{W})$$

It is enough that (W) hold with a fixed change in the fraction $1/2$. Finally assume the localized marked comparison

$$\left| \int h_i(x) K_\tau(x, a) d\mu_i(x) \right| \leq \frac{C}{m} \int H_i(x)^2 (1 + b^{-1} + H_i(x)b^{-2}) K_{c\tau}(x, a) d\mu_i(x) + m^{-B}, \quad (\text{M})$$

with the stated negligible tail bounds. Then

$$\sup_{i, x_i \in J_0} |h_i(x_i)| \leq C_{J_0, J} \sqrt{\log m/m}.$$

Proof. Suppose $D > C_*b$, and take a maximizing marked row, with coordinate x_0 , signed deviation $h_i(x_0)$, absolute deviation Δ , and boundary distance $\delta = \phi(x_0)$. Then $\Delta \leq R_m$ and

$$\delta = D/\Delta \geq C_*b/R_m, \quad \delta/b \geq C_*/R_m \rightarrow \infty.$$

The ratio grows polynomially, so Gaussian tails beyond $\delta/3$ are superpolynomially small. We only shift the test center by a fixed multiple of b , not by the potentially much larger Δ .

Choose fixed constants B_0 and $A_0 = 2B_0$, in that order, with B_0 sufficiently large for the Gaussian tails below. If $h_i(x_0) = \Delta > 0$, center the kernel at $a = x_0 + A_0b$. For $|x - a| \leq B_0b$, monotonicity gives

$$h_i(x) \geq \Delta - (x - x_0) \geq \Delta - (A_0 + B_0)b \geq \Delta/2,$$

provided C_* is large, since $\Delta = D/\delta \geq c_J C_*b$. For a negative deviation center at $a = x_0 - A_0b$ and reverse signs. All these intervals remain inside J , and their cutoff weights differ by a factor $1 + o(1)$ from δ . Throughout the larger $\delta/3$ neighborhood, weighted maximality gives $|h_i| \leq C\Delta$, and (W) gives $H_i \leq C\Delta$. By Lemma 1, the total kernel mass is at least a constant, whereas the mass outside $|x - a| \leq B_0b$ is at most $C \exp(-cB_0^2) + o(1)$. Taking B_0 large therefore yields

$$\left| \int h_i K_\tau(\cdot, a) d\mu_i \right| \geq c\Delta.$$

The far part outside the $\delta/3$ neighborhood is negligible by the preceding polynomial lower bound on δ/b . The right side of (M), using the bounded broader kernel mass, is at most

$$\frac{C\Delta^2}{m} (1 + b^{-1} + \Delta b^{-2}) + m^{-B} = o(\Delta),$$

because $\Delta \leq R_m \rightarrow 0$, $mb^2 = C_0 \log m$, and $mb \rightarrow \infty$. This contradiction proves $D \leq C_*b$. Since ϕ is bounded below on J_0 , the conclusion follows. $\square$

Tensor bookkeeping still required. For the actual marked coordinates, x_i depends on the deleted column. The full-mean deviations satisfy the exact relation $\ell u_i - x_i = (\ell/M)(mu_i - s)$. The coordinate differences between deleted columns are $O(R_m/m)$ on a compact endpoint range. Thus (W) follows from the weighted maximum of the full-mean deviations, since the maximizing row is much farther than b (and than the $O(R_m/m)$ coordinate discrepancy) from the boundary. This last conversion should be written explicitly in the application, together with spectral truncation and global-tail estimates supplying the two comparison hypotheses.